

TE Curve Verification Pipeline Official Model Verification Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Verdict

This update accepts the completed `wave 4.3` mixture-density heads campaign into the official `TE Curve Verification Pipeline` offline verification matrix.

Decision

- `wave 4.3` mixture-density heads are verified as an exploratory dispersion-aware baseline.
- No MDN candidate is promoted over the current accepted `TE Curve Verification Pipeline` leaders.
- The strongest MDN forward candidate is `track2h_mdn_k3_Fw`.
- The strongest MDN backward candidate is `track2h_mdn_k2_Bw`.
- The strongest MDN global candidate is `track2h_mdn_k2_global`.
- The best MDN backward candidate improves over the previous quantile/probabilistic and robust-loss `wave 4 series` backward branches, but it remains behind the accepted periodic temporal backward leader.
- The project continues to maintain direction-parallel `Fw`, `Bw`, and `global` surfaces instead of collapsing them into one scalar ranking.

Source Package

This official report consolidates these refreshed artifacts:

- metric matrix:
doc/reports/analysis/track2/Track 2 Directional Model Comparison.md ;
- matrix summary:
output/validation_checks/track2_reference_comparison/2026-06-13-17-24-53__track2_full_directional_family_matrix_track2h_mixture_density_heads_track2_refresh_2026_06_13/validation_summary.yaml ;
- per-condition metrics:
output/validation_checks/track2_reference_comparison/2026-06-13-17-24-53__track2_full_directional_family_matrix_track2h_mixture_density_heads_track2_refresh_2026_06_13/per_condition_metrics.csv ;
- best-model collage report:
doc/reports/analysis/track2/best_model_collage_report/[2026-06-13]/track2_best_model_collage_report.md ;
- multi-model curve comparison report:
doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-06-13]/track2_multi_model_curve_comparison_report.md ;
- operator launch logs:
output/validation_checks/track2_operator_launch_logs/2026-06-13-17-24-48_track2h_mixture_density_heads_track2_refresh_2026_06_13/ .

Candidate Refresh

The refresh added 6 registry-backed wave 4.3 MDN candidates covering two- and three-component mixture heads across the global , Fw , and Bw surfaces.

Surface	Profile	Deterministic Curve	Candidate
global	mdn_k2	mixture expectation	track2h_mdn_k2_global
Fw	mdn_k2	mixture expectation	track2h_mdn_k2_Fw
Bw	mdn_k2	mixture expectation	track2h_mdn_k2_Bw
global	mdn_k3	mixture expectation	track2h_mdn_k3_global
Fw	mdn_k3	mixture expectation	track2h_mdn_k3_Fw
Bw	mdn_k3	mixture expectation	track2h_mdn_k3_Bw

The matrix source group is track2h_mixture_density_heads_registry . The matrix now contains 153 candidates.

Verification Rules

Surface	Training or Archive Scope	Evaluation Scope
Fw	forward-only training or archive	forward TE Curve Verification Pipeline curves only

Surface	Training or Archive Scope	Evaluation Scope
Bw	backward-only training or archive	backward TE Curve Verification Pipeline curves only
global	forward and backward training together	both directions, reported by direction and combined

The MDN heads are not compared as raw mixture tensors. The official matrix evaluates only the deterministic TE curve exposed by each checkpoint: the mixture expectation computed from component logits and component means.

Current Leaders

Overall Direction Leaders

Direction	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
forward	rcim_retuned_GBM19_Fw	0.001089	0.001299	2.372	4.912
backward	periodic_gru_sequence_Bw	0.002392	0.002639	5.466	14.820
global forward	periodic_gru_sequence_global	0.002777	0.003025	6.267	13.580
global backward	periodic_gru_sequence_global	0.002630	0.002872	6.010	12.693

Repository-Owned Static Baselines

Direction	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
backward	tree_Bw	0.003258	0.003651	7.051	14.116
global forward	tree_global	0.002998	0.003364	6.590	11.601
global backward	tree_global	0.003290	0.003702	7.118	13.703

Wave 4.3 Mixture Density Result

Surface	Candidate	Profile	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
Bw	track2h_mdn_k2_Bw	mdn_k2	0.002668	0.002947	5.880	15.593
Bw	track2h_mdn_k3_Bw	mdn_k3	0.002730	0.003009	6.049	15.512
global forward	track2h_mdn_k2_global	mdn_k2	0.003263	0.003582	7.261	14.354
Fw	track2h_mdn_k3_Fw	mdn_k3	0.003226	0.003487	7.164	11.702
Fw	track2h_mdn_k2_Fw	mdn_k2	0.003329	0.003593	7.388	12.771
global forward	track2h_mdn_k3_global	mdn_k3	0.003415	0.003713	7.594	15.012
global backward	track2h_mdn_k3_global	mdn_k3	0.003701	0.004023	8.129	16.261

Surface	Candidate	Profile	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
global backward	track2h_mdn_ k2_global	mdn_k2	0.003735	0.004073	8.194	15.436

The strongest MDN `Bw` candidate improves over the best quantile/probabilistic `Bw` candidate from `0.002935` to `0.002668` TE Curve Verification Pipeline MAE. It also improves over the best robust-loss `Bw` candidate from `0.003078` to `0.002668` curve-verification MAE. This makes `track2h_mdn_k2_Bw` the strongest Wave 4 series backward surface so far.

The same improvement does not carry over to `Fw` or `global`. The strongest MDN `Fw` candidate remains behind the best probabilistic `Fw` candidate, and both MDN global candidates remain weaker than the best probabilistic global surface. The campaign-level mixture diagnostics also showed effective component counts near `1.0`, so this official refresh does not prove a stable learned multimodal TE distribution.

Visual Evidence

The `2026-06-13` collage and overlay bundles were regenerated after the MDN refresh. They include explicit Wave 4.3 mixture-density sections for `Fw`, `Bw`, and `global` collage evidence, plus explicit `Fw` and `Bw` MDN overlay sections.

The visual source-coverage validator passed with `track2h_mixture_density_heads_registry` exposed in both visual reports: `collage=6`, `overlay_forward=2`, and `overlay_backward=2`.

The visual package supports the matrix decision: MDN is a useful dispersion-aware backward branch, but it does not replace the strongest periodic temporal branch or the forward paper-reference leader.

Campaign Update Ledger

Date	Campaign or Update	Candidate Scope	Matrix Status	Visual Status	Decision
2026-06-13	Wave 4.3 mixture-density heads refresh	6 MDN <code>global</code> , <code>Fw</code> , and <code>Bw</code> candidates	included in the 153 - candidate matrix	dated collage and overlay bundles regenerated with MDN sections	verified exploratory baseline; not promoted
2026-06-12	Wave 4.2 quantile/probabilistic refresh	6 quantile and Gaussian <code>global</code> , <code>Fw</code> , and <code>Bw</code> candidates	included in the 147 - candidate matrix	dated collage and overlay bundles regenerated	verified exploratory baseline; not promoted
2026-06-11	Wave 4.1 robust-loss dispersion-aware refresh	9 robust <code>global</code> , <code>Fw</code> , and <code>Bw</code> candidates	included in the 141 - candidate matrix	dated collage and overlay bundles regenerated with Wave 4 series sections	verified exploratory baseline; not promoted
2026-06-10	Wave 3.3 curve-aware training refresh	12 pointwise-control, centered-shape, offset, and full-composite <code>global</code> , <code>Fw</code> , and <code>Bw</code> candidates	included in the 132 - candidate matrix	dated collage and overlay bundles regenerated with Wave 3.3 sections	verified exploratory baseline; not promoted
2026-06-08	Wave 3.2 harmonic-offset probe refresh	6 clean and harmonic <code>global</code> , <code>Fw</code> , and <code>Bw</code> candidates plus 3 rechecked Wave 3.1 candidates	included in the 120 - candidate matrix	dated collage and overlay bundles regenerated with Wave 3.2 sections	verified exploratory baseline; not promoted
2026-06-04	Wave 3.1 offset-aware probe refresh	3 <code>global</code> , <code>Fw</code> , and <code>Bw</code> sequential residual offset candidates	included in the 114 - candidate matrix	dated collage and overlay bundles regenerated	verified exploratory baseline; not promoted

Date	Campaign or Update	Candidate Scope	Matrix Status	Visual Status	Decision
2026-05-28	Wave 2.3 residual harmonic temporal hybrid refresh	18 global , Fw , and Bw residual harmonic GRU/LSTM candidates	included in the 111 - candidate matrix	collage and overlay reports refreshed with Wave 2.3 sections	verified exploratory baseline; not promoted over Wave 2.2 or accepted TE Curve Verification Pipeline baselines

Closeout Decision

Wave 4.3 mixture-density heads do not change the accepted TE Curve Verification Pipeline baseline. The current direction-parallel decision remains:

- Fw : rcim_retuned_GBM19_Fw remains the strongest overall forward candidate. Within the MDN package, track2h_mdn_k3_Fw is the strongest forward-only candidate, but it trails the best probabilistic forward branch.
- Bw : periodic_gru_sequence_Bw remains the strongest accepted repository-owned backward candidate. Within the MDN package, track2h_mdn_k2_Bw is the strongest backward candidate and the strongest Wave 4 series backward result so far.
- global : periodic_gru_sequence_global remains the strongest accepted bidirectional neural candidate. Within the MDN package, track2h_mdn_k2_global is the strongest global surface, but it trails the best probabilistic global candidate. The next modeling step should not be another MDN-only variant. The evidence supports moving to the first real Wave 5.1 hybrid structured campaign, while keeping robust and probabilistic losses as candidate ingredients for later multi-task / multi-head integration. A Wave 4.4 latent-state / hysteresis-aware branch remains justified if the next decision is to pursue state compensation before structured harmonic priors.